

Here's the explanation of both files:

1. Dockerfile

A `Dockerfile` is used to create a Docker image for your application.

You have two Dockerfiles:

- One for FastAPI backend
 - One for React frontend
-

Backend Dockerfile (FastAPI)

```
FROM python:3.11

WORKDIR /app

COPY requirements.txt .

RUN pip install -r requirements.txt

COPY . .

CMD ["uvicorn", "main:app", "--host", "0.0.0.0", "--port", "8000"]
```

Line-by-Line Explanation

1.

```
FROM python:3.11
```

This tells Docker:

- Use Python 3.11 as base image
- It already contains Python installed

Think of it as:

"Start from a machine that already has Python."

2.

```
WORKDIR /app
```

Creates a folder named `/app` inside container.

All next commands run inside this folder.

Equivalent to:

```
cd /app
```

3.

```
COPY requirements.txt .
```

Copies `requirements.txt` from your computer → container.

4.

```
RUN pip install -r requirements.txt
```

Installs all Python packages:

- fastapi
- uvicorn
- pymongo
- etc.

This runs during image build time.

5.

```
COPY . .
```

Copies all backend project files into container.

Example:

- main.py
- database.py
- .env

6.

```
CMD ["uvicorn", "main:app", "--host", "0.0.0.0", "--port", "8000"]
```

Starts FastAPI server when container starts.

Meaning:

Part	Meaning
main	main.py
app	FastAPI app variable
--host 0.0.0.0	Allow outside access
--port 8000	Run on port 8000

Frontend Dockerfile (React)

```
FROM node:20
WORKDIR /app
COPY package*.json ./
RUN npm install
COPY . .
EXPOSE 5173
CMD ["npm", "run", "dev", "--", "--host"]
```

Explanation

1.

```
FROM node:20
```

Uses Node.js version 20 image.

React/Vite requires Node.js.

2.

```
WORKDIR /app
```

Creates working directory inside container.

3.

```
COPY package*.json ./
```

Copies:

- package.json
 - package-lock.json
-

4.

```
RUN npm install
```

Installs all React dependencies.

5.

```
COPY . .
```

Copies all frontend files.

6.

EXPOSE 5173

Tells Docker:

React app runs on port 5173

7.

CMD ["npm", "run", "dev", "--", "--host"]

Starts Vite React server.

--host makes app accessible outside container.

2. docker-compose.yml

This file manages multiple containers together.

Instead of starting:

- MongoDB
- Backend
- Frontend

manually, Docker Compose starts all together.

Your file:

```
version: '3.9'
```

```
services:
```

```
  mongodb:
    image: mongo
    container_name: mongodb
    ports:
```

```
    - "27017:27017"

  backend:
    build: ./backend
    container_name: fastapi_backend
    ports:
      - "8000:8000"
    depends_on:
      - mongodb
    environment:
      - MONGO_URL=mongodb://mongodb:27017

  frontend:
    build: ./frontend
    container_name: react_frontend
    ports:
      - "5173:5173"
    depends_on:
      - backend
```

Explanation

version

version: '3.9'

Docker Compose file version.

services

services:

Defines containers.

You have 3 services:

1. mongodb
 2. backend
 3. frontend
-

MongoDB Service

```
mongodb:
  image: mongo
```

Uses official MongoDB image from Docker Hub.

Equivalent to:

```
docker pull mongo
```

Container Name

```
container_name: mongodb
```

Container will be named:

```
mongodb
```

instead of random name.

Ports

```
ports:
  - "27017:27017"
```

Mapping:

```
Your PC : Docker Container
27017   : 27017
```

MongoDB accessible from local machine.

Backend Service

```
backend:
  build: ./backend
```

Docker builds image using:

backend/Dockerfile

Container Name

```
container_name: fastapi_backend
```

Container name becomes:

```
fastapi_backend
```

Port Mapping

```
ports:
  - "8000:8000"
```

Maps FastAPI port.

Access via:

```
http://localhost:8000
```

depends_on

```
depends_on:
  - mongodb
```

Start MongoDB before backend.

environment

```
environment:
  - MONGO_URL=mongodb://mongodb:27017
```

Sets environment variable inside container.

Important:

Inside Docker:

```
mongodb
```


is hostname of MongoDB container.

That's why this works in Docker.

Frontend Service

```
frontend:
  build: ./frontend
```

Uses frontend Dockerfile.

Port Mapping

```
ports:
  - "5173:5173"
```

Access React app:

```
http://localhost:5173
```

depends_on

```
depends_on:
  - backend
```

Backend starts before frontend.

Overall Architecture

```
React Frontend
  ↓
FastAPI Backend
  ↓
MongoDB Database
```

Important Difference

Running Locally

Use:

```
MONGO_URL=mongodb://localhost:27017
```

because MongoDB runs on your PC.

Running with Docker Compose

Use:

```
MONGO_URL=mongodb://mongodb:27017
```

because containers communicate using service names.

Main Docker Commands

Build & Start

```
docker compose up --build
```

Stop

```
docker compose down
```

View Running Containers

```
docker ps
```

Real Benefit of Docker

Without Docker:

- Install Python

- Install Node
- Install MongoDB
- Handle versions manually

With Docker:

- Everything isolated
- Works same on every machine
- Easier deployment
- Easier team collaboration